Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/25/2007
PI Name:	Walsworth, Ronald L. Ph.D.		
Project Title:	Probing Planck Scale Physics with a 21Ne/3He Zeeman Maser		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cambridge	**State:**	MA
Zip Code:	02138	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	03/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: PI says was extended to 9/30/2007 (Ed., 5/18/07) NOTE: Received NCE to 03/31/2007 per PI (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	The 21Ne/3He Zeeman maser employs co-located ensembles of 21Ne and 3He atoms to provide sensitive differential measurements of the noble gas nuclear Zeeman splittings as a function of time, thereby greatly attenuating common-mode systematic effects such as uniform magnetic field variations. The 21Ne maser will serve as a precision magnetometer to stabilize the system's static magnetic field, while the 3He maser is used as a sensitive probe for violations of CPT and Lorentz symmetry by searching for small variations in the 3He maser frequency as the spatial orientation of the apparatus changes due to the rotation of the Earth. To optimize the frequency stability of the 21Ne/3He and/or the 129Xe/3He Zeeman maser, and to use the most stable of these masers in an improved test of CPT and Lorentz symmetry of the neutron. In the context of a general extension of the Standard Model of particle physics, the 21Ne/3He (or 129Xe/3He) maser will provide the most sensitive search to date for CPT and Lorentz violation of the neutron: better than 10^-32 GeV, an improvement of more than an order of magnitude over past experiments. This exceptional precision will offer a rare opportunity to probe physics at the Planck scale. A future space-based 21Ne/3He maser or related device could provide even greater sensitivity to violations of CPT and Lorentz symmetry, and hence to Planck-scale physics, because of isolation from dominant systematic effects associated with ground-based operation, and because of access to different positions in space-time.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report received this reporting period.
Bibliography Type:	Description: (Last Updated: 07/25/2019)